Surgical, Trauma & Perioperative ICU – Chapter Summary

Trauma • Burns • Post-op Care • Environmental

ICU / EDIC Board Review • MCQ-Oriented

1. Primary Survey & Trauma Resuscitation

- **<C>ABCDE:** catastrophic hemorrhage first (tourniquet, pressure)
- **Damage control resuscitation:** permissive hypotension SBP 80–90 (not in TBI), early blood 1:1:1, limit crystalloid, TXA ≤ 3 h, prevent hypothermia, calcium
- **Hemorrhage classes:** I < 15%, II 15–30% (tachycardia), III 30–40% (hypotension), IV > 40%
- **Shock index** > 1 → massive transfusion likely (ABC score ≥ 2)
- **eFAST** for hemoperitoneum, pneumothorax, tamponade
- **REBOA:** controversial (UK-REBOA: harm)

2. Damage Control Surgery

- **Stage 1:** abbreviated laparotomy – control bleeding & contamination (< 90 min)
- **Stage 2:** ICU – rewarm, correct coagulopathy & acidosis
- **Stage 3:** definitive repair within 24–48 h
- **Open abdomen:** negative pressure dressing; fistula risk; close early

3. Chest Trauma

- **Tension pneumothorax:** decompress (4th–5th ICS AAL)
- **Massive hemothorax:** > 1500 mL or > 200 mL/h × 2–4 h → thoracotomy
- **Flail chest:** analgesia (epidural / serratus block), NIV, surgical rib fixation
- **Pulmonary contusion:** worsens at 24–72 h; avoid fluid overload
- **Blunt aortic injury:** widened mediastinum → CTA → TEVAR; HR < 80, SBP < 100
- **Blunt cardiac injury:** troponin + ECG; arrhythmias
- **Resuscitative thoracotomy:** penetrating chest trauma with recent loss of signs of life

4. Abdominal & Pelvic Trauma

- **Unstable + positive FAST** → laparotomy
- **Stable solid organ injury:** non-operative management ± angioembolization (spleen, liver)
- **Pelvic fracture:** pelvic binder at greater trochanters → preperitoneal packing / angioembolization / external fixation
- **Hollow viscus injury:** delayed signs; seat-belt sign
- **Abdominal compartment syndrome** after massive resuscitation

5. Spine & Extremity

- **Cervical spine clearance:** NEXUS / Canadian C-spine; CT in obtunded
- **Compartment syndrome:** pain on passive stretch; pressure > 30 or ΔP (DBP − compartment) < 30 → fasciotomy
- **Fat embolism:** 24–72 h after long-bone fracture; Gurd criteria; early fixation ↓ risk
- **Crush syndrome:** rhabdomyolysis, hyperK → fluids before extrication

6. Burns – Assessment

- **TBSA:** rule of 9s (adults), Lund–Browder (children); palm ≈ 1%
- **Inhalation injury:** enclosed space, singed hair, carbonaceous sputum, hoarseness, stridor → early intubation (large ETT ≥ 8)
- **CO:** COHb, normal SpO_2 → 100% O_2 / HBO
- **Cyanide:** lactate > 8 → hydroxocobalamin 5 g
- **Circumferential full-thickness:** escharotomy (chest / limb)
- **Electrical burns:** deep injury, rhabdo, arrhythmias (ECG monitoring)

7. Burns – Management

- **Parkland:** 4 mL/kg × %TBSA (Ringer's lactate) in 24 h – half in first 8 h (from time of burn)
- **Modified Brooke** 2 mL/kg/%TBSA as starting point; titrate to **UO 0.5 mL/kg/h** (1 mL/kg/h electrical)
- **Fluid creep** → ACS, ARDS, compartment syndrome → albumin rescue
- **Early excision & grafting** (< 72 h)
- **Hypermetabolism:** ↑ energy & protein (1.5–2 g/kg); **oxandrolone**, **propranolol**
- **Succinylcholine** contraindicated > 24–48 h
- **Infection:** Pseudomonas, MRSA, Candida; no prophylactic antibiotics
- **Thermoregulation:** room 28–32 °C

8. Postoperative Complications

- **Anastomotic leak:** day 3–7; fever, tachycardia, ileus → CT, drain / relaparotomy
- **Postoperative AKI**, ileus, delirium, PONV
- **Postoperative pulmonary complications:** atelectasis, pneumonia; lung-protective ventilation intra-op
- **Myocardial injury after noncardiac surgery (MINS):** troponin rise, ↑ mortality
- **Bleeding:** clinical + Hb drop; return to theatre
- **VTE prophylaxis** resume early
- **Malignant hyperthermia:** ↑ $EtCO_2$, rigidity, hyperthermia, ↑ K → **dantrolene 2.5 mg/kg**, stop volatiles / sux

9. Cardiac & Thoracic Surgery

- **Post-pneumonectomy pulmonary edema:** avoid fluid overload; right-sided worse
- **Bronchopleural fistula:** persistent air leak
- **Post-cardiotomy:** tamponade, vasoplegia, LCOS, AF, bleeding → see cardio chapter
- **Esophagectomy:** anastomotic leak, chylothorax, aspiration

10. Analgesia & Regional Techniques

- **Multimodal:** paracetamol, NSAIDs (caution AKI), ketamine, lidocaine infusion, dexmedetomidine
- **Epidural:** ↓ pulmonary complications; hypotension; check coagulation (ASRA timings)
- **Regional blocks:** serratus anterior, erector spinae, TAP
- **LAST:** perioral numbness, seizures, arrhythmia → **lipid emulsion 20% 1.5 mL/kg**
- **Opioid-sparing** strategies ↓ ileus & delirium

11. Perioperative Medicine

- **Stress-dose steroids:** hydrocortisone 50–100 mg if on chronic steroids
- **Anticoagulation bridging:** generally not needed for AF (BRIDGE)
- **Perioperative β-blocker:** don't start acutely (POISE: ↑ stroke, mortality)
- **Glucose** < 180 mg/dL
- **ERAS:** early feeding, mobilization, goal-directed fluids
- **Frailty** predicts poor outcome

12. Necrotizing Soft Tissue Infection

- Pain out of proportion, crepitus, bullae, rapid spread; **LRINEC** (limited sensitivity)
- **Urgent surgical debridement** (repeat 24–48 h)
- **Antibiotics:** broad (piperacillin-tazobactam / carbapenem) + **clindamycin** (toxin suppression) ± MRSA cover
- **Fournier's gangrene:** perineum
- IVIG for streptococcal TSS (selected)

13. Environmental Emergencies

- **Hypothermia:** mild 32–35, moderate 28–32, severe < 28 °C; J waves; VF risk; ECLS rewarming if arrest; "not dead until warm and dead"; max 3 shocks < 30 °C, withhold adrenaline < 30 °C
- **Heat stroke:** > 40 °C + CNS dysfunction → rapid cooling (cold water immersion); dantrolene not effective
- **Drowning:** hypoxemia, ARDS; no difference fresh vs salt water
- **Avalanche / altitude:** HAPE (nifedipine, descent), HACE (dexamethasone)
- **Snakebite:** antivenom

14. High-Yield MCQ Pearls

- [x] Catastrophic hemorrhage before airway
- [x] Permissive hypotension 80–90 (not TBI)
- [x] TXA ≤ 3 h in trauma (CRASH-2)
- [x] Massive hemothorax > 1500 mL → thoracotomy
- [x] Pelvic binder at greater trochanters
- [x] Parkland 4 mL/kg/%TBSA, half in 8 h
- [x] Burns UO target 0.5 mL/kg/h (1 electrical)
- [x] Inhalation injury → early intubation
- [x] Sux contraindicated in burns > 24–48 h
- [x] Malignant hyperthermia → dantrolene 2.5 mg/kg
- [x] LAST → lipid emulsion 20%
- [x] Necrotizing fasciitis → surgery + clindamycin

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